clay deposits, the organic material is most significant in maintaining a reducing potential which restricts actinide migration and provides a sorption source of the mobilized fraction.

22.9. THE OKLO REACTOR

Analysis of the Oklo natural reactors (§19.10) indicates that they must have lasted for 100 000 to 500 000 years with criticality occurring periodically. They probably would have consumed about twelve metric tons of ^{235}U, releasing a total energy of 2–3 GWy at a probable power level of ~10 kW. About 1.0–1.5 t of ^{239}Pu were formed by neutron capture in ^{238}U, but the half-life of ^{239}Pu has resulted in its total decay to ^{235}U. Since a few samples enriched in ^{235}U (presumably due to this decay) have been found, it is believed that in some places in Oklo breeding conditions may have temporarily existed. The decaying fission products are estimated to have had 10^{28} β,γ disintegrations over the operating time of the reactors. The average energy release in the reactor zone was ~50 W m^{-2} which is several times greater than that planned for geological nuclear waste repositories. As a consequence, it is estimated that the fluids in the inclusions of the mineral grains in the Oklo reactors had temperatures of 450 to 600°C, well above those anticipated in deep waste repositories. There is evidence of connectively driven circulation of the fluids for distances of 30 m from the main reactor zones as well as significant dissolution and modification of minerals due to the thermal and radiation conditions. Redistribution of some elements resulted from the convective flow of the hot liquids.

The uranium minerals appear to have remained relatively stable despite this heating process. The uranium and the lanthanide elements show evidence of some small degree of localized redistribution, but were mostly retained within the reactor zone. By contrast, fission product rare gases, halogens, molybdenum and the alkali and alkaline earth elements migrated significant distances from the reactor zones. In general, it seems that elemental redistribution took place over a period of 0.5 to million years while the area was thermally hot (during and after nuclear criticality). It has been estimated that as much as a total of 10^{12} liter of hot water flowed through the reactor zones. The water leaving the reactor zones had 5×10^{-3} g U m^{-3} and ~10^{-10} M concentrations of Tc, Ru, and Nd. The Tc and Ru were oxidized to TcO_4^- and RuO_4^-, and these soluble oxyanions moved with the water as it flowed from the area, creating significant (25–35%) deficiencies of these elements in the reactor zones. The Nd was less soluble and, apparently, migrated much less during this hot period. For Tc and Ru, the migration rate seems to have been ~10^{-5} m y^{-1} in water moving at a flow of 5 m y^{-1}.

A very important observation is the evidence in Oklo that the plutonium produced by the reactors did not move during its lifetime from the site of its formation.

In summary, essentially 100% of the Pu, 85−100% of the Nd, 75−90% of the Ru and 60−85% of the Tc were retained within the reactor zones. The migrating fission products were held within a few tens of meters of these zones. Thermodynamic calculations of the temperature dependent solubilities indicate that the loss of fissiogenic elements is diffusion controlled, whereas, retention in the surrounding rocks is due to temperature dependent deposition from an aqueous solution.

While the conditions at Oklo differ in a number of aspects from those expected in nuclear repository sites, they frequently were much less favorable to retention of the radionuclides. The lack of migration of the actinides and the much slower release of Tc agree with the predictions of laboratory studies and indicate their value in validating the safety of nuclear repositories.

22.10. PERFORMANCE ASSESSMENTS OF WASTE REPOSITORIES

In this Chapter we have presented some information on studies of the environmental behavior of some fission products and actinides. We have shown how laboratory data can be used in speciation calculations to predict solubilities, etc. The knowledge gained from studies of natural analog sites and, very importantly, from the Oklo natural reactor site, has been reviewed. All of these types of studies and data are of use in assessing the probable ability of nuclear waste repositories to protect the public over the time needed required for the radioactivity to decay. This time is largely dependent on used nuclear fuel handling as discussed previously and thsu the time required may vary between some thousends of years for a full transmutation cycle to more than 100 000 years for the once through cycle. The most common design goal for these repositories is that no radioactivity shall be released within the next 1 000 years, and that the leakage after that time would be so small that it presents no danger to living species. The Environmental Protection Agency (EPA), USA, standard requires that no more than 1 000 cancers shall be caused in 10 000 years by radionuclides released from the waste, using the ICRP predictions of radiation dose effects (Ch. 15).

In the case of direct disposal of used nuclear fuel, the initiated radioactive inventory consists of actinides, fission products and activation products. As noted previously, (Ch. 21) the shorter lived fission products, such as ^{90}Sr and ^{137}Cs, and transuranic elements, such as ^{238}Pu, ^{241}Pu, ^{244}Cm, are the main contributors to the radioactivity. However, performance assessments strongly indicate that the waste form matrix and the near field engineered barriers (e.g. clay backfill, etc.), can successfully retain and prevent any migration to the far field environment for one thousand years and probably much longer ($>10^4$ years). After the first thousand years the long lived nuclides such as ^{129}I, ^{135}Cs, ^{126}Sn, ^{99}Tc and ^{79}Se among the fission products, ^{226}Ra, and the actinides ^{234}U, ^{236}U, ^{237}Np, ^{239}Pu, ^{240}Pu, and ^{241}Am become the major concern.